

Decorrelated spectral regression: An unsupervised dimension reduction method under data selection bias

Xiuqi Huang, Haotian Ni, Tingjin Luo, Hong Tao, Chenping Hou *

College of Sciences, National University of Defense Technology, Changsha 410073, China

ARTICLE INFO

Article history:

Received 7 January 2023

Revised 18 April 2023

Accepted 29 May 2023

Available online 7 June 2023

Communicated by Zidong Wang

Keywords:

Dimensionality reduction

Causal inference

Graph embedding

ABSTRACT

In manipulating high-dimensional data arising from real applications, dimension reduction methods have been proven to be effective and necessary. Nevertheless, the widespread selection bias, induced in the process of data collection, has long been unaware and its negative effects have been ignored. The accidental spurious correlation between the features brought by selection bias will deteriorate the performance of dimension reduction approaches. It is vital to remove such bias and consider its influence on dimension reduction. In this paper, we propose a novel decorrelated spectral regression algorithm for unsupervised dimension reduction on data with sample selection bias. In this method, the sample weights of the global feature distribution are learned by introducing the decorrelation regularizer. The learned weights are then combined with the regression model to obtain the dimension reduction mapping. Thus, the dimension reduction results are less affected by the accidental spurious correlation of data caused by selection bias. In addition, we also propose the updating rules of the parameters in our algorithm. A large number of experimental results on real-world datasets show that our method has achieved a significant improvement, indicating that it is effective and necessary to remove the accidental spurious correlation between features caused by selection bias in dimension reduction.

© 2023 Elsevier B.V. All rights reserved.

1. Introduction

The availability of vast amounts of information has led to an increase in the prevalence of high-dimensional data, which offer valuable insights for data analysis. Unfortunately, such data often contain redundancy and strong correlations, leading to what is commonly known as the “curse of dimensionality”. This poses significant challenges both in terms of storage capacity and computational efficiency, hindering the success of many advanced algorithms. To address this issue, researchers have increasingly turned to dimensionality reduction techniques. These methods aim to transform the original data while preserving its intrinsic structure, resulting in increased density of samples in a lower dimensional subspace, reduced storage requirements, and improved implementability. Numerous dimensionality reduction techniques have been proposed based on various assumptions. These techniques have been extensively validated on diverse datasets and have found widespread applications in fields such as hyperspectral imaging, visualization, classification, and face recognition.

However, it is important to note that in real-world scenarios, biased data collection processes can result in unexpected spurious correlations between features, leading to negative impacts on the dimensionality reduction results beyond the assumed factors [1,2]. For instance, consider a dataset of cat and dog images in which most of the dogs are photographed against a grass background while the cats appear on a variety of different backgrounds. As a consequence, the features of the dogs become highly correlated with the grass background, while those of the cats display less correlation. This additional correlation between dog features and grassland features is an unstable spurious correlation, resulting from data selection bias. If not addressed, the negative impact of this spurious correlation can extend to the dimensionality reduction results and ultimately affect the performance of the learning model [3,4], thereby increasing the likelihood of misjudgments on the data after dimensionality reduction.

Despite the widespread prevalence of data selection bias, the impact of resulting spurious correlations in dimension reduction has been underestimated. Current dimensionality reduction methods often overlook this bias [5,6], making it challenging to obtain accurate results when dealing with biased data. Therefore, it is necessary to devise a dimensionality reduction algorithm capable of eliminating additional correlation between features in the data. However, traditional algorithms encounter several obstacles:

* Corresponding author.

E-mail addresses: xiuqi_huang@nudt.edu.cn (X. Huang), nihotian17@163.com (H. Ni), luotingjin03@nudt.edu.cn (T. Luo), taohong.nudt@hotmail.com (H. Tao), hcpnudt@hotmail.com (C. Hou).

(1) How to measure the selection bias in the data to further investigate the possible additional correlation? (2) How to remove the accidental correlation in high dimensional data? (3) How to make the removal of correlation beneficial to the result of dimension reduction?

This paper proposes a novel approach called the Decorrelated Spectral Regression (DSR) model for unsupervised dimension reduction of data affected by selection bias. The traditional graph embedding method is utilized to construct the adjacency matrix, and the response vector is generated. In the subsequent regression step, a decorrelation regularization term is introduced to remove correlation between target features and remaining features. The distribution of remaining features is balanced by learning global sample weight, and the spectral regression loss is weighted accordingly. This unifies the weighted regression model and the decorrelation regularization term into a structure that reduces the impact of spurious correlation between features on dimension reduction results. Additionally, we present an effective update iteration rule to update the model's parameters. In summary, our contributions can be summarized as follows:

- We investigate an important but rarely studied problem, how to reduce dimensionality of data affected by selection bias. Accidental correlation between features caused by biased selection of data is prevalent in practical applications, despite rare discussion on this aspect.
- To address this issue, we propose a novel decorrelated regularized regression model for unsupervised dimension reduction of data affected by selection bias. This model reduces unexpected bias by introducing a decorrelation regularizer into the regression model that learns dimension reduction mapping. Additionally, we provide effective update rules to optimize the parameters and variables of the Decorrelated Spectral Regression (DSR) algorithm.
- Through numerous comprehensive experiments, we demonstrate the effectiveness and superiority of our method. Compared to other comparison methods, our proposed approach significantly enhances dimension reduction performance on datasets affected by selection bias.

This paper is organized as follows. Section 2 provides a review of related work, followed by an introduction to notation and preliminary knowledge in Section 3. In Section 4, we present details on our algorithm with relevant analysis. In Section 5, we conducted comprehensive experiments on several datasets and provided parameter sensitivity analysis and convergence analysis. Finally, we conclude this paper in Section 6.

2. Related Work

This section describes the established relevant research, including dimensionality reduction and causal inference.

Dimensionality reduction techniques are crucial in addressing challenges presented by high-dimensional data, including storage and computational efficiency. Both linear and nonlinear approaches have been developed to extract an optimal transformation while still maintaining the intrinsic structure of the original datasets. While linear techniques like PCA [7,8] and LDA [9] use covariance matrix information to identify the maximum variance direction or retain the most discriminative information during projection, respectively, nonlinear methods leverage the manifold hypothesis, such as LLE [10], LPP [11], ISOMAP [12], and LE [13]. Moreover, kernel-based methods extend linear algorithms to nonlinear ones, including KPCA [14] and KDA [15]. Recently, graph-based techniques using the graph-embedding framework have

gained popularity in constructing an affinity graph to determine performance [16]. To preserve desired geometric properties in the original space, various methods like NPE [17], SPP [18], UDP [19], MFA [20] capture both local and global information.

Causal inference provides a cogent and effective tool for further analysis of the interaction between factors, especially the estimation of causal effects when confounders and bias exist. Rosenbaum and Rubin [21] proposed that propensity score matching and inverse probability weighting could be used to achieve the balance of confounding variables. This far-reaching and influential method has been widely applied in various fields such as economics [22], social sciences [23] and epidemiology [24]. In recent years, some literature have attempted to balance the confounder distribution by directly optimizing the sample weight. Hainmueller uses the specified sample moment to directly adjust the sample weight and puts forward the entropy balance [25]. Athey et al. proposed an approximate residual balance algorithm to learn sample weights through Lasso residual regression [26]. Kuang et al. proposed a data-driven differentiated confounder balancing algorithm to balance distributions by learning different weights of confounding factors [27]. These methods provide some effective methods to estimate the causal effect, however there are certain limitations on the number of variables and the causal prior knowledge.

3. Preliminaries

3.1. Related Notation

Throughout this paper, matrices are written as capital letters and vectors are denoted as lowercase letters. $X \in \mathbb{R}^{n \times d}$ is the data matrix, n is the sample size, and the dimension of the feature is denoted by d . c is the feature dimensions after dimension reduction. y is the responses generated by the solving eigenproblem. a is the transformation vector.

3.2. Graph Embedding View of Dimension Reduction

In this section, we briefly describe the dimension reduction framework from the graph embedding perspective [16].

We assume that for n samples $\{x_i\}_{i=1}^n \in \mathbb{R}^d$, $X = [x_1, \dots, x_n]^T$, the graph G with n vertices can construct by data X . Each vertex in graph G corresponds to a sample point, and the relationship between samples can be reflected by the edges in the graph. Such a graph G can be represented by a symmetric matrix $n \times n$, where the element W_{ij} represents the existence and weight of edges between points i and j .

In this way, the graph G and the adjacency matrix W can reveal some properties of the data sets. The task of graph embedding is to represent each vertex of the graph as a low-dimensional vector under certain constraints, which usually include preserving the similarity between point pairs measured by edge weights.

For the result of embedding mapping $y = [y_1, y_2, \dots, y_n]^T$, the most common constraint is to ensure that when vertex i and vertex j are 'close', the embedding mapping result y_i and y_j are also 'close', that is, the appropriate embedding result should be optimized through the following formula:

$$\sum_{i,j=1}^n (y_i - y_j)^2 W_{ij} \quad (1)$$

According to the references [16], after some algebraic transformation, we can get the optimal y corresponding to the solution when the eigenvalue of the following eigenvalue decomposition problem is taken to the maximum:

$$Wy = \lambda Dy \quad (2)$$

where W is the adjacency matrix of graph G , D is a diagonal matrix, and $D_{ii} = \sum_j W_{ji}$.

The calculation method of y presented here is based on the information given in the data set, and it is necessary to learn a dimension reduction mapping applicable to the whole data set.

When we choose a linear mapping model, i.e., $y_i = f(x_i) = a^T x_i$, Eq. (2) can be converted to:

$$XWX^T a = \lambda XDX^T a. \quad (3)$$

where a is the embedded mapping, which is equal to the corresponding eigenvector of the maximum eigenvalue of the equation. And the solution for a can also be converted into a regression form [28].

The above expression can also be called as the linear extension of graph embedding, which can be extended into LDA, LPP, NPE, and other forms depending on the choice of W [28–30].

4. Decorrelated Spectral Regression

4.1. Problem and Definition

As previously discussed, datasets generated through unevenly biased selection may contain additional spurious correlations that can harm algorithmic models. One way to mitigate data selection bias is by adding extra samples to the dataset, such as images of dogs with different backgrounds in the previous example, to weaken the correlation between dogs and grass backgrounds. However, collecting additional data can be challenging due to cost and other constraints. Alternatively, we can adjust the weight of each sample to mitigate the harm caused by selection bias.

The problem of selection bias can be reframed using causal inference tools. The bias between sample distribution, represented by the H , will impact the distribution of both input features X and mapping results Z . In causal inference terminology, H is referred to as a confounder, which affects both X and Z . Fig. 1 provides an abstract summary of the relationship between H , X , and Z . Here, X corresponds to the input feature, and Z is the mapping result.

The presence of a confounding variable can obscure important information regarding dimension reduction ($X \rightarrow Z$). To improve algorithm performance, it is necessary to eliminate the influence of the confounder ($H \rightarrow X$ and $H \rightarrow Z$). The confounding effect of H is reflected in the biased distribution of samples in the dataset, which can be rectified by balancing the sample weights. To facilitate further analysis, we introduce the following definition. Without loss of generality, we assume that all features and labels are binary (other types can be converted to binary through binning and one-hot encoding), which aids in discussion and comprehension.

Definition 1 (Remaining Features). When the j -th feature of X is selected as the target feature (i.e., $\mathbf{X}_j$), the remaining features are defined as follows: $\mathbf{X}_{-j} = \mathbf{X} \setminus \mathbf{X}_j$, and $\mathbf{X}_{-j}$ is generated by replacing the j -th column of $\mathbf{X}$ with 0.

Definition 2 (Treated Group and Control Group). After the target feature is given, treated group and control group are divided according to the value of the target feature of the sample. For the i -th sample, if the j -th feature is 1, this sample is classified as treated group, otherwise it is classified as control group. That means that treated group $TG_j = \{\mathbf{X}_i | \mathbf{X}_{ij} = 1\}$, and control group $CG_j = \{\mathbf{X}_i | \mathbf{X}_{ij} = 0\}$.

According to Definition 1, we divide the data matrix into two parts: the target feature and the remaining features. The remaining features are obtained by replacing the column of the target feature

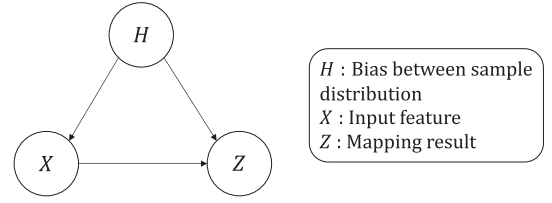


Fig. 1. Overview of the proposed causal graph explaining the causal effect of the data selection bias.

with zero. Definition 2 specifies that samples with a target feature of 1 belong to the treated group, while those with a target feature of 0 belong to the control group. Once the target feature is selected, the samples can be divided into corresponding groups based on the value of the target feature. By comparing the distribution of the treated and control groups, we can determine the correlation between the target feature and the remaining features. To reduce the additional correlation, we adjust the distribution to achieve equilibrium. [3].

4.2. Decorrelation Regularizer

In this section, we will gradually introduce the decorrelation regularizer and derive a solution method for determining global sample weights that balance the distribution. To achieve a balanced sample distribution, it is essential to identify a suitable metric that quantifies the deviation between the sample distribution, which is also illustrated as H in Fig. 1.

In this paper, we adopt moments as a primary tool for measuring deviation, which are widely acknowledged as effective in characterizing distributions. According to reference [27], a distribution can be determined when the moment of each order has been provided. And our study primarily focuses on the first moment, as it provides sufficient information while considering the prior binary assumption.

The moment bias. We first compute the first moments of the treated group and control group. The first-order moment of the remaining features in treated group TG_j is:

$$\tilde{\mathbf{X}}_{-j} = \frac{\mathbf{X}_{-j}^T \cdot \mathbf{X}_j}{\mathbf{1}_n^T \cdot \mathbf{X}_j} \quad (4)$$

where $\mathbf{1}_n = [1, 1, \dots, 1] \in \mathbb{R}^{n \times 1}$, $\mathbf{X}_j$ is targeted features and $\mathbf{X}_{-j}$ is the remaining features.

For the remaining features in control group CG_j , the first-order moment is:

$$\hat{\mathbf{X}}_{-j} = \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{1}_n - \mathbf{X}_j)}{\mathbf{1}_n^T \cdot (\mathbf{1}_n - \mathbf{X}_j)} \quad (5)$$

In the binary case, the distribution difference between the treated group TG_j and the control groups CG_j mentioned above can be expressed as the inequality between the first moments of the treated group TG_j and the control groups CG_j . Therefore, we introduce the single moment bias B_s under single feature, which can be defined in the following form:

$$B_s = \left\| \frac{\mathbf{X}_{-j}^T \cdot \mathbf{X}_j}{\mathbf{1}_n^T \cdot \mathbf{X}_j} - \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{1}_n - \mathbf{X}_j)}{\mathbf{1}_n^T \cdot (\mathbf{1}_n - \mathbf{X}_j)} \right\|_2^2 \quad (6)$$

The single feature moment bias defined here is obtained after we select the j -th feature as the targeted feature, and it can be used to measure the degree of distribution inconsistency between the treated group TG_j and the control groups CG_j under j -th feature,

and the targeted feature j is the confounding variable mentioned in our earlier analysis.

Since we do not have enough prior knowledge about the causal structure relationship between the data, a feasible method for data $X \in \mathbb{R}^{n \times d}$ is to set all d features as target features in turn and analyze the distribution of the corresponding remaining features.

By combining the moment bias B_s for each feature, we can naturally obtain the global moment bias B_g , which is defined as follows:

$$B_g = \sum_{j=1}^d \left\| \frac{\mathbf{X}_{-j}^T \cdot \mathbf{X}_j}{\mathbf{1}_n^T \cdot \mathbf{X}_j} - \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{1}_n - \mathbf{X}_j)}{\mathbf{1}_n^T \cdot (\mathbf{1}_n - \mathbf{X}_j)} \right\|_2^2 \quad (7)$$

Decorrelation regularizer. To eliminate this bias, we can introduce sample weight $\mathbf{u} \in \mathbb{R}^{n \times 1}$ and assign different weights to each sample. According to the previous definition, we can obtain the adjusted first moments of treated group $\tilde{\mathbf{X}}_{-j}^u$ and the adjusted first moments of treated group $\tilde{\mathbf{X}}_{-j}^u$:

$$\tilde{\mathbf{X}}_{-j}^u = \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot \mathbf{X}_j)}{\mathbf{u}^T \cdot \mathbf{X}_j} \quad (8)$$

$$\hat{\mathbf{X}}_{-j}^u = \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot (\mathbf{1}_n - \mathbf{X}_j))}{\mathbf{u}^T \cdot (\mathbf{1}_n - \mathbf{X}_j)} \quad (9)$$

where $\odot$ refers to the Hadamard product.

Similar to the definition of single moment bias, we can define the adjusted single moment bias B_s^u as follow:

$$B_s^u = \left\| \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot \mathbf{X}_j)}{\mathbf{u}^T \cdot \mathbf{X}_j} - \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot (\mathbf{1}_n - \mathbf{X}_j))}{\mathbf{u}^T \cdot (\mathbf{1}_n - \mathbf{X}_j)} \right\|_2^2 \quad (10)$$

As for the adjusted single moment bias B_s^u , $\mathbf{u} \in \mathbb{R}^{n \times 1}$ can completely balance the moment bias for single feature. For global features, we need to learn $\mathbf{u}^* \in \mathbb{R}^{n \times d}$ to fully balance global bias, which is extremely difficult in the case of high dimensions. Note that for different d features, the adjusted samples are the same, and the same weight can be used for samples with different target features. In this way, we can get the adjusted global moment bias B_g^u :

$$B_g^u = \sum_{j=1}^d \left\| \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot \mathbf{X}_j)}{\mathbf{u}^T \cdot \mathbf{X}_j} - \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot (\mathbf{1}_n - \mathbf{X}_j))}{\mathbf{u}^T \cdot (\mathbf{1}_n - \mathbf{X}_j)} \right\|_2^2 \quad (11)$$

When the moment distribution bias of the global feature adjusted by sample weight $\mathbf{u} \in \mathbb{R}^{n \times 1}$ decreases, the additional correlation between the target feature and the remaining feature is weakened, and the influence derived from the data selection bias is also removed to a certain extent.

And sample weights $\mathbf{u} \in \mathbb{R}^{n \times 1}$ can be learned by:

$$\mathbf{u} = \arg \min_{\mathbf{u}} \sum_{j=1}^d \left\| \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot \mathbf{X}_j)}{\mathbf{u}^T \cdot \mathbf{X}_j} - \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot (\mathbf{1}_n - \mathbf{X}_j))}{\mathbf{u}^T \cdot (\mathbf{1}_n - \mathbf{X}_j)} \right\|_2^2 \quad (12)$$

By learning sample weights from the moment bias, we can balance the distribution of the treated group TG_j and control groups CG_j by weighting different samples. By utilizing such approach, not only is the spurious correlation between the target feature and the remaining features weakened, the confounding factor that impacts both the input and outcome is also effectively balanced. Additionally, it helps eliminate the influence of data selection bias to some extent. In other words, by balancing the distribution of sample weights learned, we can consider that the influence of $H \rightarrow X$ and $H \rightarrow Z$ in Fig. 1 have been weakened.

4.3. The Framework of DSR

Based on obtaining the decorrelation regularizer, we can combine it with the dimension reduction method of graph embedding perspective. After obtaining the constructed graph information and response vector, we jointly optimized the decorrelation regularizer and the regression model for solving the embedded mapping, so that the embedded results could better reflect the properties of the data.

The general framework of the DSR algorithm proposed by us can be summarized as follows:

Constructing the graph G and choosing the edge weights matrix W : For data $X \in \mathbb{R}^{n \times d}$, we use it to construct a graph G with n vertices, each vertex corresponds to the sample point. And under unsupervised conditions, the edge connection is based on the following principles: Select the appropriate number of neighbors p , if x_j is one of the p closest points to x_i , then an edge is connected between x_j and x_i .

For the obtained graph G , the adjacency matrix W can be obtained according to the information of edges in the graph G . W is a symmetric matrix, and its element W_{ij} contains the information of the graph:

- (a) $W_{ij} = 0$, if there is no edge between vertices i and j .
- (b) $W_{ij} = s(i, j)$, if there is an edge between vertices i and j .

The element $s(i, j)$ is used to measure the similarity between point i and point j , which usually has two forms: Heat kernel:

$$s(i, j) = e^{-\frac{\|x_i - x_j\|^2}{\sigma^2}} \text{ and Simple-minded: } s(i, j) = 1.$$

Generating responses y : Find $y_0, y_1, \dots, y_c$, the largest $c + 1$ generalized eigenvectors of eigenproblem:

$$W\mathbf{y} = \lambda D\mathbf{y} \quad (13)$$

where D is a diagonal matrix, and $D_{ii} = \sum_j W_{ji}$. It is straightforward to show that the first eigenvector is a vector of all ones with eigenvalue 1. And let $Y = [y_1, y_2, \dots, y_n]$, $Y \in \mathbb{R}^{n \times c}$.

Decorrelated Spectral Regression: In spectral regression model [28], the embedding mapping A is learned based on the original data X and response Y . However, the existence of selection bias may make the model performance worse. Since the decorrelation regularizer can learn sample weight $\mathbf{u}$ to balance the confounder, we use the learned sample weight $\mathbf{u}$ to adjust the regression loss, and combine to optimize the adjusted regression loss and the decorrelation regularization term. The objective function of this step can be summarized as follows:

$$\begin{aligned} \min_{\mathbf{u}, A} & \| (X \cdot A - Y) \odot (\mathbf{u} \cdot \mathbf{1}_c^T) \|_2^2 \\ \text{s.t.} \quad & B_g^u \leq \lambda_1, \mathbf{u} \geq 0, \|\mathbf{u}\|_2^2 \leq \lambda_2, \|A\|_2^2 \leq \lambda_3, \sum_{i=1}^n (\mathbf{u}_i - 1)^2 \leq \lambda_4 \end{aligned} \quad (14)$$

where B_g^u is global decorrelation regularizer, and $B_g^u = \sum_{j=1}^d \left\| \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot \mathbf{X}_j)}{\mathbf{u}^T \cdot \mathbf{X}_j} - \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot (\mathbf{1}_n - \mathbf{X}_j))}{\mathbf{u}^T \cdot (\mathbf{1}_n - \mathbf{X}_j)} \right\|_2^2$, $\mathbf{u}$ is the sample weight, Y is the response, and A is the embedding mapping.

The global decorrelation regularizer B_g^u aims to reduce unexpected extra correlation. The term $\mathbf{u} \geq 0$ guarantees that each sample weight is non-negative. Using the term $\|\mathbf{u}\|_2^2 \leq \lambda_2$ can reduce the variance and make the model more stable. The $\lambda_3 \geq \|A\|_2^2$ is used to control the amounts of shrinkage. The term $\sum_{i=1}^n (\mathbf{u}_i - 1)^2 \leq \lambda_4$ avoids the occurrence of weights that are all zero.

Embedding for Dimension Reduction: For transformation matrix $A \in \mathbb{R}^{d \times c}$, data $X \in \mathbb{R}^{n \times d}$ can be embedded into c dimensional subspace, embedding result Z is calculated by following equation:

$$Z = X \cdot A \quad (15)$$

4.4. Optimization Algorithm

In this subsection, we decompose the optimization problem of Eq. (14) into two subproblems and perform iterative optimization. The specific optimization details are as follows.

According to the previous analysis, the Eq. (14) is equivalent to the minimum of $J(A, \mathbf{u})$:

$$J(A, \mathbf{u}) = \|(X \cdot A - Y) \cdot (\mathbf{u} \cdot \mathbf{1}_c^T)^{1/2}\|_F^2 + \lambda_1 \sum_{j=1}^d \left\| \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot \mathbf{X}_j)}{\mathbf{u}^T \cdot \mathbf{X}_j} \right\|_2^2 - \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot (\mathbf{1}_n - \mathbf{X}_j))}{\mathbf{u}^T \cdot (\mathbf{1}_n - \mathbf{X}_j)} \Big\|_2^2 + \lambda_2 \|\mathbf{u}\|_2^2 + \lambda_3 \|A\|_2^2 + \lambda_4 \left(\sum_{i=1}^n \mathbf{u}_i - 1 \right)^2, \quad (16)$$

$$s.t. \quad \mathbf{u} \geq 0,$$

To optimize Eq. (16), we update A and $\mathbf{u}$ iteratively:

A-subproblem: When updating A in Eq. (16) with $\mathbf{u}$ being fixed, the objective function that needs to be optimized is converted to:

$$J(A) = \|(X \cdot A - Y) \cdot (\mathbf{u} \cdot \mathbf{1}_c^T)^{1/2}\|_F^2 + \lambda_3 \|A\|_2^2 \quad (17)$$

which is a form of regression. The partial derivative of $J(A)$ with respect to A can then be computed as follows.

$$\frac{\partial J(A)}{\partial A} = 2 \cdot X^T \cdot (X \cdot A - Y) \cdot C \cdot C^T + 2 \cdot \lambda_3 \cdot A \quad (18)$$

$$\text{where } C = (\mathbf{u} \cdot \mathbf{1}_c^T)^{1/2}.$$

u-subproblem: When updating $\mathbf{u}$ and fixing A in Eq. (16), the objective function we need to optimize is:

$$J(\mathbf{u}) = \|(X \cdot A - Y) \cdot (\mathbf{u} \cdot \mathbf{1}_c^T)^{1/2}\|_F^2 + \lambda_1 \sum_{j=1}^d \left\| \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot \mathbf{X}_j)}{\mathbf{u}^T \cdot \mathbf{X}_j} \right\|_2^2 - \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{u} \odot (\mathbf{1}_n - \mathbf{X}_j))}{\mathbf{u}^T \cdot (\mathbf{1}_n - \mathbf{X}_j)} \Big\|_2^2 + \lambda_2 \|\mathbf{u}\|_2^2 + \lambda_3 \left(\sum_{i=1}^n \mathbf{u}_i - 1 \right)^2, \quad (19)$$

$$s.t. \quad \mathbf{u} \geq 0,$$

To ensure that the weights $\mathbf{u}$ are non-negative during the iteration, we set $\mathbf{u} = \mathbf{v} \odot \mathbf{v}$, where $\mathbf{v} \in \mathbb{R}^{n \times 1}$. Then Eq. (19) can be rewritten as follows:

$$J(\mathbf{v}) = \|(X \cdot A - Y) \cdot ((\mathbf{v} \odot \mathbf{v}) \cdot \mathbf{1}_c^T)^{1/2}\|_F^2 + \lambda_1 \sum_{j=1}^d \left\| \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{v} \odot \mathbf{v} \odot \mathbf{X}_j)}{\mathbf{v} \odot \mathbf{v}^T \cdot \mathbf{X}_j} - \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{v} \odot \mathbf{v} \odot (\mathbf{1}_n - \mathbf{X}_j))}{(\mathbf{v} \odot \mathbf{v})^T \cdot (\mathbf{1}_n - \mathbf{X}_j)} \right\|_2^2 + \lambda_2 \|\mathbf{v} \odot \mathbf{v}\|_2^2 + \lambda_3 \left(\sum_{i=1}^n \mathbf{v}_i \odot \mathbf{v}_i - 1 \right)^2 \quad (20)$$

The partial gradient of $J(\mathbf{v})$ with respect to $\mathbf{v}$ is calculated as follows:

$$\begin{aligned} \frac{\partial J(\mathbf{v})}{\partial \mathbf{v}} &= (\mathbf{1}_c^T \cdot ((X \cdot A - Y)^T \odot (X \cdot A - Y)^T))^T \odot \mathbf{v} \\ &+ \lambda_1 \sum_{j=1}^d 4 \cdot \left(\frac{\partial J_b}{\partial \mathbf{v}} \odot (\mathbf{1}_d \cdot \mathbf{v}^T) \right)^T \cdot J_b \\ &+ 4 \cdot \lambda_2 \cdot \mathbf{v} \odot \mathbf{v} \odot \mathbf{v} + 4 \cdot \lambda_3 \left(\sum_{i=1}^n \mathbf{v}_i \odot \mathbf{v}_i - 1 \right) \cdot \mathbf{v}, \end{aligned} \quad (21)$$

where

$$J_b = \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{v} \odot \mathbf{v} \odot \mathbf{X}_j)}{(\mathbf{v} \odot \mathbf{v})^T \cdot \mathbf{X}_j} - \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{v} \odot \mathbf{v} \odot (\mathbf{1}_n - \mathbf{X}_j))}{(\mathbf{v} \odot \mathbf{v})^T \cdot (\mathbf{1}_n - \mathbf{X}_j)} \quad (22)$$

$$\begin{aligned} \frac{\partial J_b}{\partial \mathbf{v}} &= \frac{\mathbf{X}_{-j}^T \odot (\mathbf{X}_j \cdot \mathbf{1}_d^T) \cdot ((\mathbf{v} \odot \mathbf{v})^T \cdot \mathbf{X}_j)}{((\mathbf{v} \odot \mathbf{v})^T \cdot \mathbf{X}_j)^2} \\ &- \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{v} \odot \mathbf{v} \odot \mathbf{X}_j)^T \cdot \mathbf{X}_j^T}{((\mathbf{v} \odot \mathbf{v})^T \cdot \mathbf{X}_j)^2} \\ &- \frac{\mathbf{X}_{-j}^T \odot ((\mathbf{1}_n - \mathbf{X}_j) \cdot \mathbf{1}_d^T) \cdot ((\mathbf{v} \odot \mathbf{v})^T \cdot (\mathbf{1}_n - \mathbf{X}_j))}{((\mathbf{v} \odot \mathbf{v})^T \cdot (\mathbf{1}_n - \mathbf{X}_j))^2} \\ &+ \frac{\mathbf{X}_{-j}^T \cdot (\mathbf{v} \odot \mathbf{v} \odot (\mathbf{1}_n - \mathbf{X}_j)) \cdot (\mathbf{1}_n - \mathbf{X}_j)^T}{((\mathbf{v} \odot \mathbf{v})^T \cdot (\mathbf{1}_n - \mathbf{X}_j))^2} \end{aligned} \quad (23)$$

After updating $\mathbf{v}$ by gradient descent, we can get $\mathbf{u}^{(t)}$ at t -th iteration:

$$\mathbf{u}^{(t)} = \mathbf{v}^{(t)} \odot \mathbf{v}^{(t)} \quad (24)$$

Before the convergence of objective function Eq. (14), we will continue to iteratively update A and $\mathbf{u}$. From the above update rules, we can see that our solving model is jointly affected by the decorrelation regularizer and the regression model. This form of joint optimization can remove the accidental correlation between features and facilitate the performance of the regression model.

Based on the above analyses, the framework of DSR is summarized in the Algorithm. 1.

Algorithm 1: Decorrelated Spectral Regression (DSR)

Input: $X \in \mathbb{R}^{n \times d}$, parameters $\lambda_1, \lambda_2, \lambda_3, \lambda_4$, the number of Reduced dimension c and the number of nearest neighbors p

Output: subspace $Z \in \mathbb{R}^{n \times c}$

- 1: Construct feature adjacency matrix W and calculate the diagonal matrix D
 - 2: Generate Responses Y by Eq. (13)
 - 3: Initialize the sample weight $\mathbf{u} \leftarrow \mathbf{1}_n$
 - repeat**
 - 4: update A Fixing $\mathbf{u}$, compute A from A-subproblem in Eq. (18)
 - 5: update $\mathbf{u}$ Fixing A , compute $\mathbf{u}$ from u-subproblem in Eq. (21)
 - until** converges or max iteration is reached
 - 6: Return embedding mapping A^*
 - 7: Get embedding outcome $Z \leftarrow X \cdot A^*$
-

4.5. Complexity Analysis

The computation of DSR mainly involves two parts. For graph construction, responses generation, and embedding, the complexity is $O(n^2(d + \log n) + ncp)$, where c is the feature dimensions after dimension reduction and p is the number of nearest neighbors to construct the graph. During the production of optimization, the major cost is to compute the loss $J(A, \mathbf{u})$, update the global sample weights $\mathbf{u}$, and embedding mapping A , and the complexity of each iteration is $O(nd^2)$, where n is the sample size and d is the dimension of data.

4.6. Theoretical Analysis

In this section, we conduct a theoretical analysis of our algorithm. Since sample weight $\mathbf{u}$ plays an important role in the solution of dimension reduction mapping, our analysis will focus on the properties of sample weight $\mathbf{u}$. When solving the sample weight $\mathbf{u}$, a key assumption needs to be satisfied.

Assumption 1 (Overlap, [26]). For any variable $X_{\cdot j}$, if it setting as the treatment variable, it has $\forall j, 0 < P(X_{\cdot j} = 1 | X_{\cdot, -j}) < 1$.

Lemma 4.1 [31]. If X are binary, and $\forall j, 0 < P(X_{\cdot j} = 1 | X_{\cdot, -j}) < 1$, then $\forall i, 0 < P(X_i = x) < 1$, where X_i is i -th row in X .

A lot of the work related with treatment effect estimation is based on the above assumption [26]. For binary data satisfying the overlap assumption, Lemma 4.1 proved by reference [31] guarantees that there exists $0 < P(X_i = x) < 1$, meaning that for every given partition based on the target feature, the treated group and control group exist simultaneously.

Theorem 1. Let $X \in \mathbb{R}^{n \times d}$. If the conditions of Lemma 4.1 satisfied and the feature size d is finite, then $\exists \mathbf{u}$ such that

$$\lim_{n \rightarrow \infty} \sum_{j=1}^d B_g^u = 0 \quad (25)$$

with probability 1.

Proof. Since $\|\cdot\| \geq 0$, Eq. (25) is equivalent to $\forall j, \forall k \neq j$

$$\lim_{n \rightarrow \infty} \left(\frac{\sum_{i: X_{i,j}=1, X_{i,k}=1} \mathbf{u}_i}{\sum_{i: X_{i,j}=1} \mathbf{u}_i} - \frac{\sum_{i: X_{i,k}=1, X_{i,j}=0} \mathbf{u}_i}{\sum_{i: X_{i,k}=1} \mathbf{u}_i} \right) = 0 \quad (26)$$

with probability 1.

For $\mathbf{u}^*$, from Lemma 4.1 and Law of Large Number, $0 < P(X_i = x) < 1, \forall x, \forall i, t = 1$ or 0 ,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i: X_{i,j}=t} \mathbf{u}_i^* = \lim_{n \rightarrow \infty} \sum_{x: X_{i,j}=t} \frac{1}{n} \sum_{i: X_i=x} \frac{1}{P(X_i=x)} = 2^{d-1} \quad (27)$$

Since the analysis in this paper is based on binary features, the following equation holds:

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i: X_{i,k}=1, X_{i,j}=1} \mathbf{u}_i^* &= 2^{d-2} \\ \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i: X_{i,j}=0} \mathbf{u}_i^* &= 2^{d-1}, \\ \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i: X_{i,k}=1, X_{i,j}=0} \mathbf{u}_i^* &= 2^{d-2} \end{aligned} \quad (28)$$

Therefore, the following equation holds with probability 1:

$$\lim_{n \rightarrow \infty} \sum_{j=1}^d B_g^u = \frac{2^{d-2}}{2^{d-1}} - \frac{2^{d-2}}{2^{d-1}} = 0. \quad (29)$$

Lemma 4.2. [31] If $0 < P(X_i = x) < 1$ for all x , there exists a solution $\mathbf{u}^*$ satisfies Eq. 25 equals 0 and variables in $\mathbf{X}$ are independent after balancing by $\mathbf{u}^*$.

According to Theorem 1, it is known that for a binary dataset satisfying the overlap assumption, there exists a sample weight that satisfies Eq. 25. Combining with Lemma 4.2, the correlation

Table 1

Summary of the benchmark data sets.

Datasets	sample	feature	class
Amazon	958	800	10
Caltech	1123	800	10
DSLR	157	800	10
Webcam	295	800	10
OH1	640	800	10
OH2	678	800	10
OH3	681	800	10
USPS	1800	256	10

between variables after balancing with such sample weights is reduced and tends towards independence. This shows that sample weight $\mathbf{u}^*$ is learnable and advantageous for improving our results.

5. Experiments

In this section, we present a series of experiments to demonstrate the effectiveness of our proposed DSR algorithm. There is no unified standard for evaluating the effectiveness of dimensionality reduction, and a mainstream approach is to evaluate it based on the performance of learning models on the reduced data. The better the performance on the reduced data, the more effective the dimensionality reduction algorithm is considered to be, as it contains more useful information. In this paper, we utilize the performance of K-means algorithm on the reduced data to measure the quality of the dimensionality reduction method, which is also one of the most widely used evaluation methods. We commence by introducing the dataset, comparative methods, experimental settings, and evaluation metrics. Subsequently, we furnish a comprehensive analysis of the experimental results, encompassing a comparison with other algorithms, a sensitivity analysis of the algorithm's parameters, and convergence tests.

5.1. Datasets

Since our analysis is based on the scenario under the data selection bias, we mainly consider this characteristic when selecting the datasets. The data set used in our experiment mainly includes three sources, Office-Caltech¹, Office-Home² and USPS³. These data sets are publicly available and their information is outlined below: (Table. 1).

- The Office-Caltech dataset is a widely used image dataset in transfer learning for object recognition. It consists of four labeled subsets, namely Amazon, Caltech, DSLR and Webcam, each of which corresponds to different image sources and collecting methods. These subsets contain pictures of 10 objects categories with diverse backgrounds. We utilized the SURF and Bag-of-words techniques on publicly available datasets for processing. [32,33]
- The Office-Home dataset is another commonly used object recognition dataset in transfer learning, consisting of hundreds of images of common objects found in home and office settings. We used data from three randomly divided subsets (OH1, OH2, OH3), each containing hundreds of pictures of 10 categories. Data processing was performed using similar techniques as those used for the Office-Caltech subset, and these three subsets are also publicly available. [34,35]

¹ <https://github.com/jindongwang/transferlearning/tree/master/data>

² <https://github.com/BUPT-GAMMA/IJCAI2020-DCKM/tree/master/data>

³ <https://cs.nyu.edu/~roweis/data.html>

- The USPS dataset is a commonly used handwriting recognition dataset, consisting of grayscale images of handwritten numbers ranging from 0 to 9, originally from the United States Postal Service. We utilized a subset of this dataset that is also publicly available. [36]

5.2. Compared Methods

We compare our DSR algorithm with the baseline and the following unsupervised dimension reduction algorithms.

Table 2

Summary of Normalized Mutual Information Results on 8 Datasets for 7 Methods (NMI $\pm$ STD%).

Dataset	Method						
	Baseline	PCA	SR	LPP	ISOMAP	UAE	DSR
Amazon	40.87 $\pm$ 1.93	44.99 $\pm$ 1.50	44.43 $\pm$ 1.56	35.92 $\pm$ 1.57	34.36 $\pm$ 2.39	44.60 $\pm$ 1.31	45.32 $\pm$ 1.39
Caltech	20.27 $\pm$ 0.74	23.49 $\pm$ 1.24	22.54 $\pm$ 0.89	18.41 $\pm$ 1.25	17.27 $\pm$ 1.27	23.63 $\pm$ 0.86	23.39 $\pm$ 1.67
DSLR	48.21 $\pm$ 3.43	51.61 $\pm$ 3.07	51.67 $\pm$ 2.94	49.57 $\pm$ 2.17	44.79 $\pm$ 2.75	51.14 $\pm$ 2.47	52.28 $\pm$ 2.51
Webcam	35.75 $\pm$ 4.54	46.33 $\pm$ 3.08	46.76 $\pm$ 2.82	33.81 $\pm$ 2.40	30.70 $\pm$ 3.29	46.70 $\pm$ 1.70	55.27 $\pm$ 1.73
OH1	33.46 $\pm$ 1.81	36.62 $\pm$ 1.76	36.43 $\pm$ 1.85	30.58 $\pm$ 2.03	30.21 $\pm$ 2.19	40.35 $\pm$ 2.17	41.01 $\pm$ 2.23
OH2	31.30 $\pm$ 2.11	34.12 $\pm$ 1.76	33.80 $\pm$ 1.66	33.31 $\pm$ 2.05	29.51 $\pm$ 2.32	37.41 $\pm$ 1.28	38.71 $\pm$ 1.77
OH3	20.86 $\pm$ 2.22	24.77 $\pm$ 1.35	24.84 $\pm$ 1.34	23.30 $\pm$ 1.20	19.64 $\pm$ 2.05	29.82 $\pm$ 1.29	35.73 $\pm$ 1.86
USPS	51.04 $\pm$ 0.98	51.33 $\pm$ 1.17	51.75 $\pm$ 1.01	55.78 $\pm$ 1.28	53.03 $\pm$ 1.53	53.53 $\pm$ 1.92	57.25 $\pm$ 2.28

Table 3

Summary of Adjusted Rand Index Results on 8 Datasets for 7 Methods (ARI $\pm$ STD%).

	Baseline	PCA	SR	LPP	ISOMAP	UAE	DSR
Amazon	19.65 $\pm$ 2.11	25.86 $\pm$ 2.57	25.23 $\pm$ 2.52	12.37 $\pm$ 1.41	10.74 $\pm$ 1.96	26.73 $\pm$ 1.68	26.97 $\pm$ 2.19
Caltech	8.21 $\pm$ 0.85	12.37 $\pm$ 1.30	10.36 $\pm$ 0.90	6.34 $\pm$ 0.67	5.46 $\pm$ 0.95	11.54 $\pm$ 0.93	13.21 $\pm$ 0.98
DSLR	20.85 $\pm$ 3.33	27.03 $\pm$ 4.17	26.77 $\pm$ 4.40	21.72 $\pm$ 4.40	16.71 $\pm$ 3.40	27.25 $\pm$ 3.23	27.92 $\pm$ 2.36
Webcam	10.91 $\pm$ 4.08	22.55 $\pm$ 4.10	21.88 $\pm$ 3.83	10.86 $\pm$ 2.44	7.91 $\pm$ 2.93	23.64 $\pm$ 2.00	29.76 $\pm$ 2.14
OH1	15.25 $\pm$ 1.48	18.55 $\pm$ 1.71	18.56 $\pm$ 2.15	10.71 $\pm$ 2.06	10.05 $\pm$ 2.27	20.12 $\pm$ 1.78	25.94 $\pm$ 1.91
OH2	12.87 $\pm$ 2.31	17.86 $\pm$ 2.33	17.56 $\pm$ 2.05	10.65 $\pm$ 1.82	9.61 $\pm$ 1.95	20.92 $\pm$ 2.08	23.69 $\pm$ 2.32
OH3	7.67 $\pm$ 2.17	11.10 $\pm$ 1.40	11.16 $\pm$ 1.31	6.21 $\pm$ 1.06	5.34 $\pm$ 1.01	14.25 $\pm$ 1.16	24.03 $\pm$ 3.20
USPS	37.69 $\pm$ 2.53	38.74 $\pm$ 2.92	38.81 $\pm$ 2.78	38.11 $\pm$ 3.50	39.37 $\pm$ 3.06	40.22 $\pm$ 2.36	51.46 $\pm$ 3.25

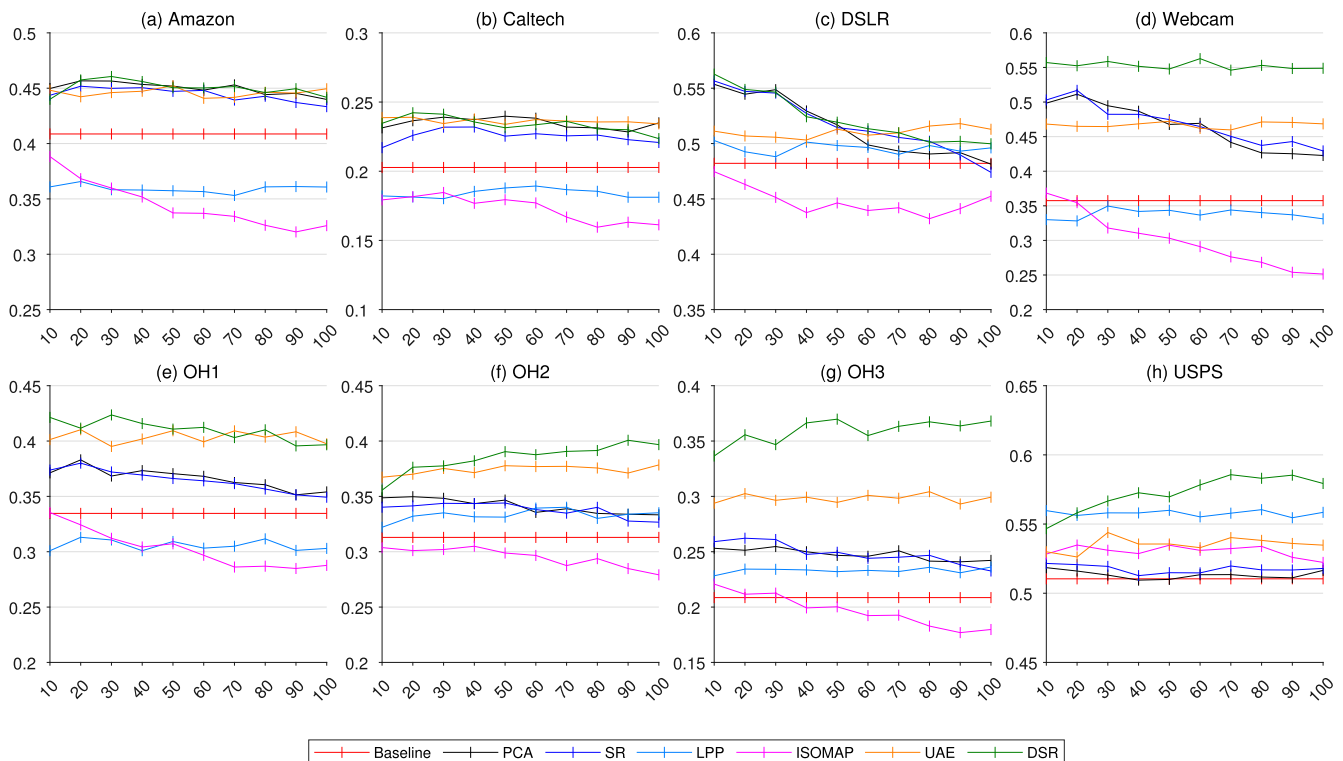


Fig. 2. Summary of Normalized Mutual Information Results for 7 Methods on 8 Datasets with Different Numbers of Reduced Dimensionality.

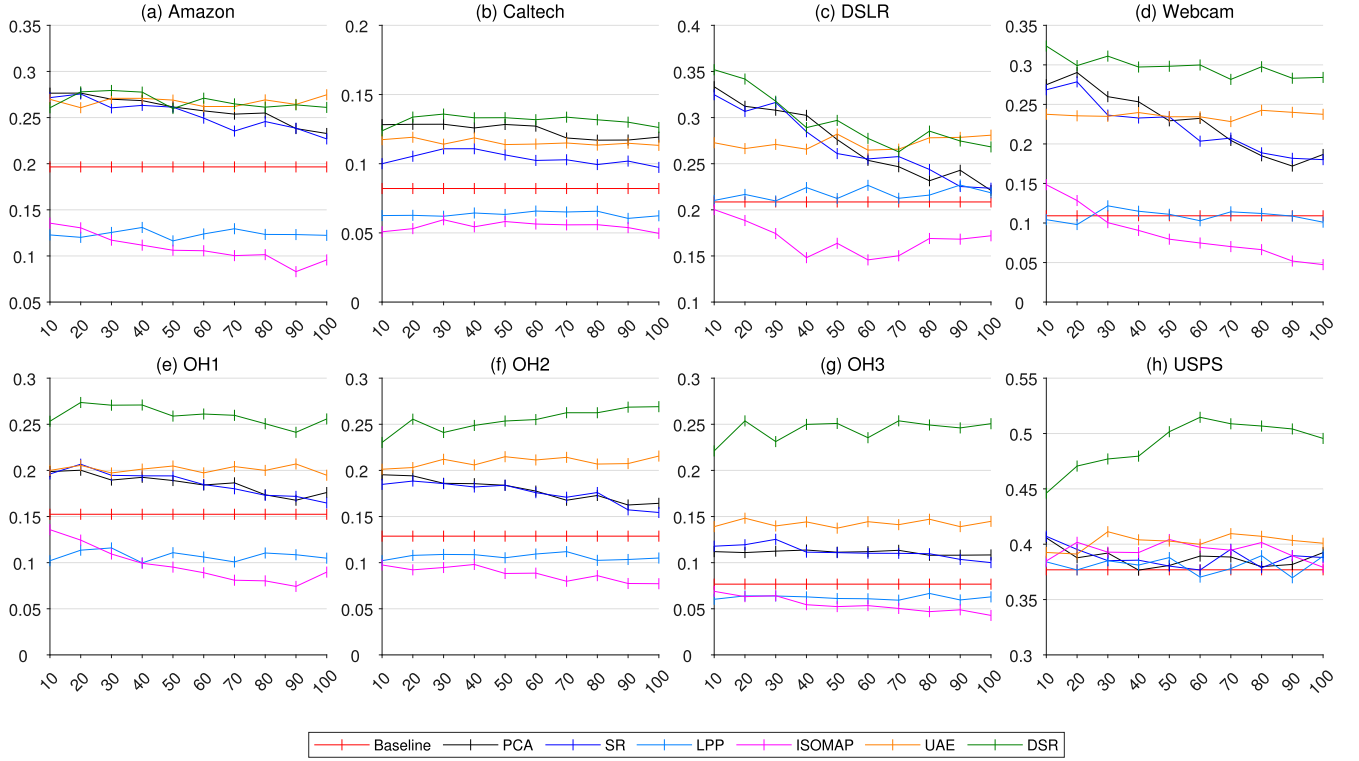


Fig. 3. Summary of Adjusted Rand Index Results for 7 Methods on 8 Datasets with Different Numbers of Reduced Dimensionality.

5. **ISOMAP**: Isometric mapping [37]: ISOMAP introduced “geodesic distance” to solve the problem that Euclidean distance cannot cope with nonlinear manifolds.
6. **UAE**: Unsupervised Adaptive Embedding [38]: an unsupervised adaptive embedding for dimensionality reduction, which is a linear method based on graph embedding and preserves local information and global geometric structure of the original data.

5.3. Experiment Setup

We compare the methods by performing K-means clustering on the dimension reduction results, where the number of central clusters is determined by the number of classes in the datasets. For our DSR algorithm, we fix $\lambda_4 = 1$ and search for the optimal values of λ_1, λ_2 , and λ_3 . Specifically, we search for λ_1 from the set $10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2$, and search for λ_2 and λ_3 from the set $10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3$. For each method requiring graph construction, we search for and utilize the optimal nearest neighbors number. For Baseline, we perform clustering directly on all features. For UAE, we adopted a similar approach to the original paper for data preprocessing and selecting the value of r , and searched for the best result among the values of $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2\}$ for the λ parameter. For the number of reduced dimensions, we choose it from the set $\{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$. To reduce the experimental error, we repeat the experiment 20 times and apply NMI and ARI to measure the clustering effect.

5.4. Evaluation Metrics

In the analysis of experimental results, the clustering results on the data after dimension reduction were used to evaluate the performance of the method. The Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI) were selected to evaluate the clus-

tering effect. The value ranges of evaluation metrics NMI and ARI are $[0, 1]$ and $[-1, 1]$ respectively. Meanwhile, for these two metrics, the higher the value, the more consistent the results are with the real situation.

The Normalized Mutual Information (NMI) can be calculated by the following formula:

$$NMI(A, B) = \frac{I(A, B)}{\sqrt{H(A)H(B)}} \quad (30)$$

For random variables A and B , $H(A)$ and $H(B)$ represent the entropy of A and B , respectively. $I(A, B)$ corresponds to the mutual information of random variables A and B .

The formula of the Adjusted Rand Index (ARI) can be described as follows:

$$ARI = \frac{RI - E[RI]}{\max(RI) - E(RI)} \quad (31)$$

In the formula above RI is the Rand Index, and $RI = \frac{a+b}{C_n^2}$, where a represents the number of instance pairs divided into the same cluster in the actual partitioning and clustering results, b represents the number of instance pairs divided into different clusters in the actual partitioning and clustering results, and C_n^2 is equal to $\frac{n(n-1)}{2}$.

5.5. Result Analysis

The experimental results of different methods on various datasets, as evaluated by the NMI and ARI metrics, are respectively presented in Table 2 and Table 3. The method with the highest average performance for each dataset is highlighted in bold within the tables. Furthermore, we conducted t-tests on the experimental results to compare the significance of the differences in means between the optimal and suboptimal methods listed in the table for each dataset. At a significance level of 0.05, the following results were obtained: For the experiments under the NMI metric,

except for the Caltech dataset, it can be concluded at a significance level of 0.05 that the mean of the best-performing method is significantly different from the mean of the second-best method on the remaining seven datasets. Therefore, it can be inferred that our proposed method outperforms other methods on these seven datasets. Under the ARI metric, it can be inferred at a significance level of 0.05 that the mean of the best-performing method is significantly different from the mean of the second-best method on all datasets except for the Amazon and DSLR datasets. Thus, our proposed method is deemed superior to other methods on these six datasets.

Furthermore, we investigated the relationship between the experimental results under different metrics and the variation of dimensions, which are respectively presented in Fig. 2 and Fig. 3. The results of our experiments indicate that the proposed method demonstrates competitiveness and provide us with the following observations.

- Our method demonstrates superior performance compared to existing methods on most datasets, as confirmed by statistical analysis using two different metrics, NMI and ARI.
- Specifically, our approach demonstrates a significant improvement in comparison to the SR method, indicating the crucial role of joint decorrelation terms in the optimization process.

- Our method's superiority over other methods is effectively demonstrated in certain datasets, where significant improvements were observed. The variability of biases across different datasets may account for the differing levels of improvement.

5.6. Parameters Sensitivity

In this section, we investigated the impact of parameters in our model on its performance. For our proposed Decorrelated Spectral Regression (DSR) algorithm, there are three important parameters λ_1, λ_2 and λ_3 . The parameter λ_1 was used to adjust the scale of the decorrelation regularizer, parameters λ_2 and λ_3 acted on the regularizer of sample weight $\mathbf{u}$ and $\mathbf{A}$ respectively to prevent the overfitting of the reconstruction function. We study the influence of parameters in different dimensions, where the search range of the reduced dimension c is $\{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$, the search range of parameter λ_1 is $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2\}$, and the search range of parameter λ_2 and λ_3 is $\{10^{-2}, 10^{-1}, 1, 10, 10^2\}$. When we study one of the parameters, we keep the other parameters fixed, and the experimental results under different evaluation indexes are shown in Fig. 4 and Fig. 5. The results of the above experiments convey that our method's performance is not sensitive to λ_1, λ_2 , or λ_3 over a wide range of parameter values.

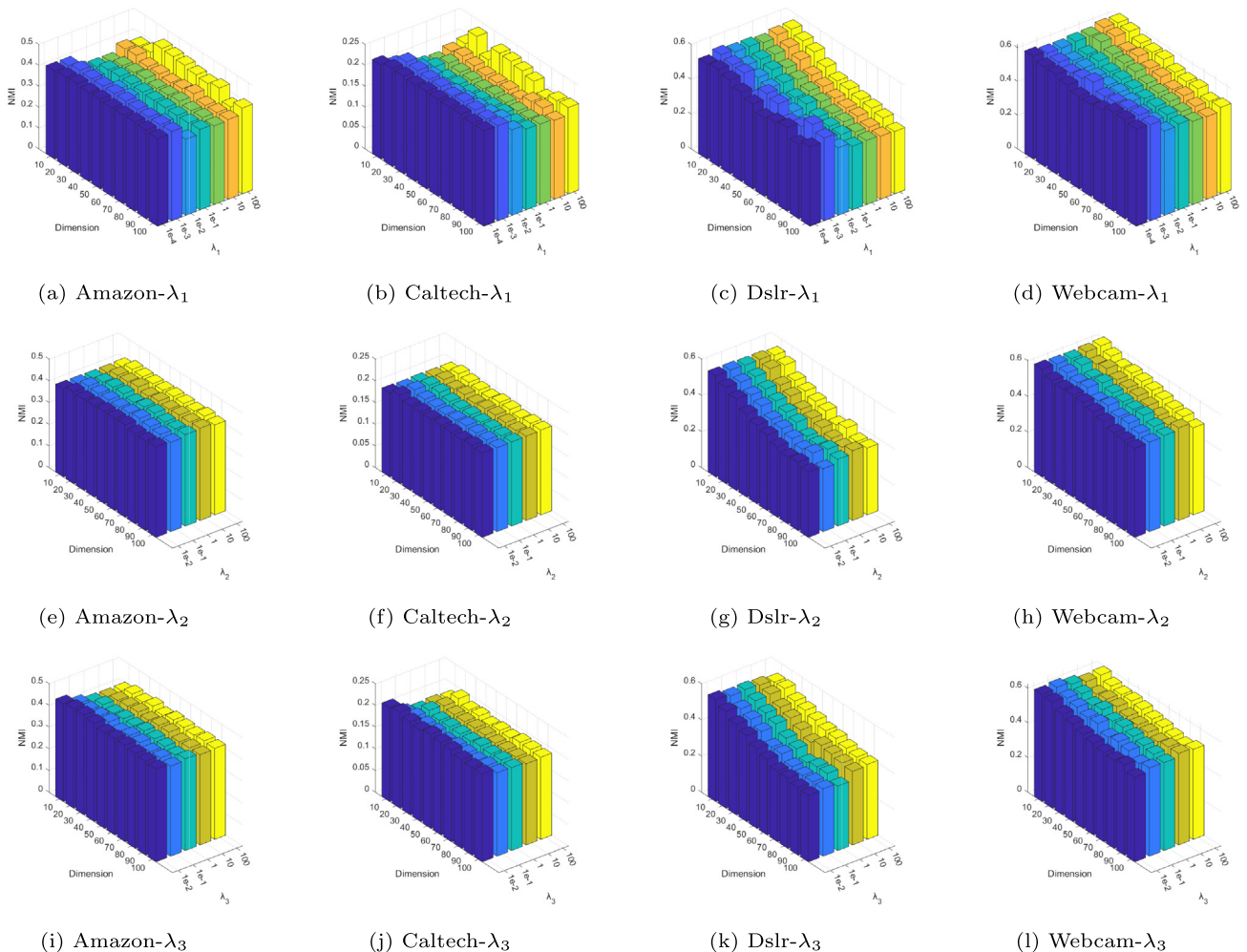


Fig. 4. Parameter sensitivity analysis with respect to λ and dimension under NMI index.

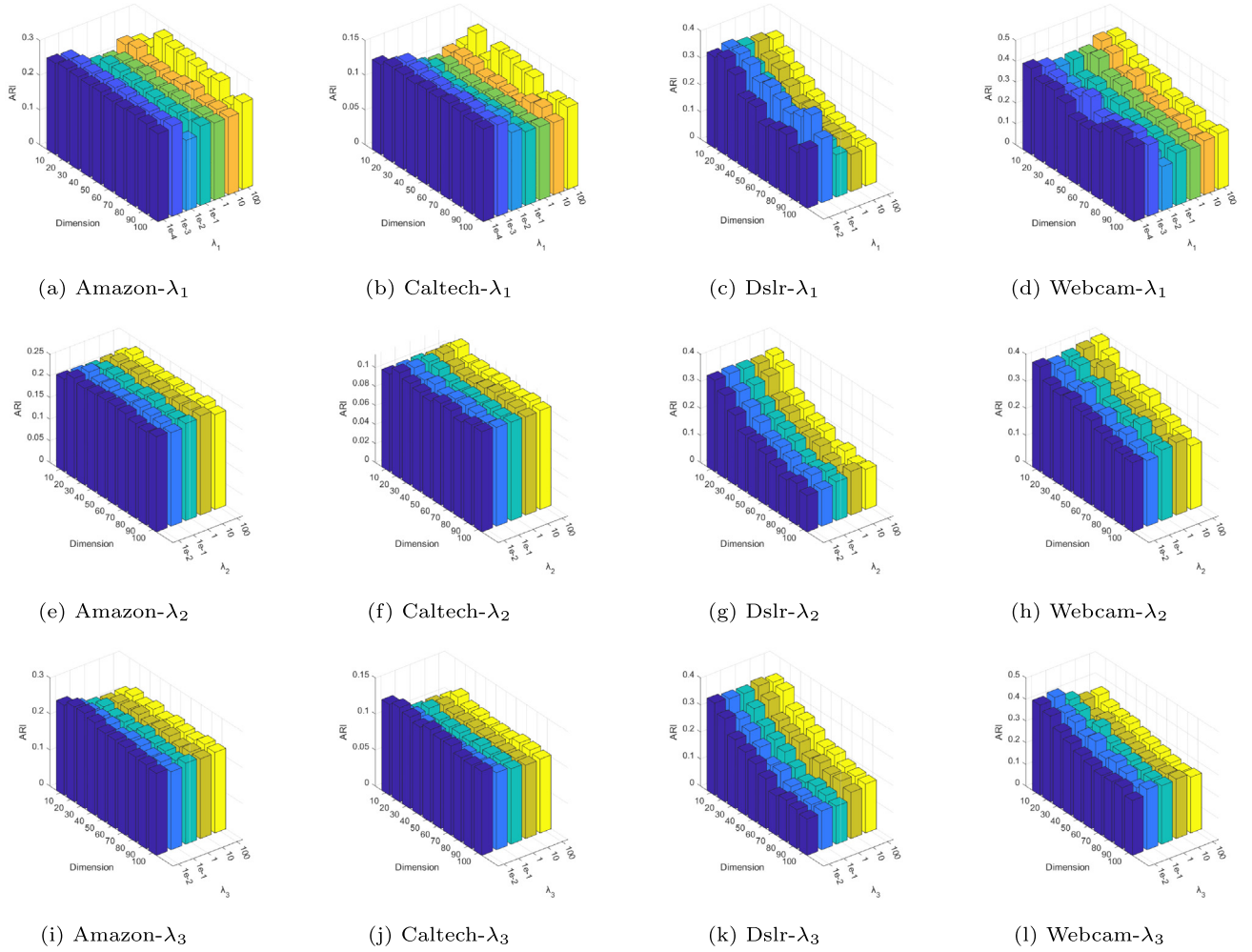


Fig. 5. Parameter sensitivity analysis with respect to λ and dimension under ARI index.

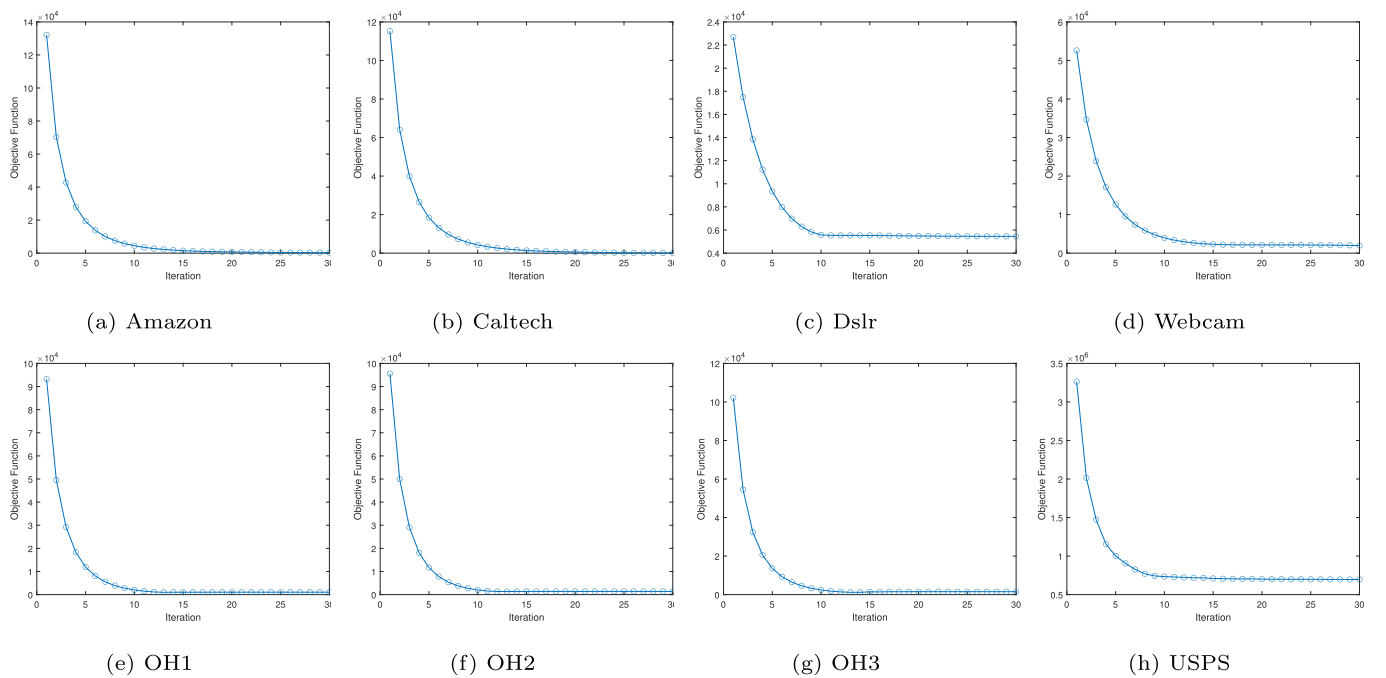


Fig. 6. The convergence curves of the objective function on different datasets.

5.7. Convergence Test

In this section, we verify the convergence of the objective function during optimization. We recorded the changes in the value of the objective function over different data sets as the iteration progressed, and the relevant experimental results are shown in Fig. 6. From the experimental results, we can see that the value of the objective function decreases with the increase of the number of iterations and converges after finite iterations. We can set 30 iterations as the maximum number of iterations. And from the experiment above, the convergence of our algorithm is verified.

6. Conclusion

In this paper, we explore a rarely studied but important problem: dimension reduction method with data selection bias. In view of the unexpected correlation between features caused by selection bias and the possible unsatisfactory dimension reduction results, we propose a novel Decorrelated Spectral Regression (DSR) algorithm for unsupervised dimension reduction on data affected by selection bias. In this method, the regularizer used to remove the unexpected correlation between features is integrated into the graph embedding dimension reduction framework, and the sample weight and dimension reduction mapping used to eliminate the bias are jointly optimized. For the optimization of DSR, we propose an effective iteration strategy to obtain the optimal solution. A large number of comprehensive experimental results demonstrate the effectiveness of our method.

CRediT authorship contribution statement

Xiuqi Huang: Methodology, Methodology, Investigation, Formal analysis, Writing - original draft, Writing - review & editing. **Haotian Ni:** Methodology, Software, Investigation, Resources. **Tingjin Luo:** Resources, Writing - review & editing. **Hong Tao:** Resources, Writing - review & editing. **Chenping Hou:** Conceptualization, Methodology, Supervision, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was partially supported by the Key NSF of China under Grant No. 62136005, the NSF of China under Grant No. 61922087.

References

- [1] J.J. Heckman, Sample selection bias as a specification error, *Econometrica: Journal of the econometric society* (1979) 153–161.
- [2] J. Huang, A. Gretton, K. Borgwardt, B. Schölkopf, A. Smola, Correcting sample selection bias by unlabeled data, *Advances in neural information processing systems* 19 (2006).
- [3] X. Wang, S. Fan, K. Kuang, C. Shi, J. Liu, B. Wang, Decorrelated clustering with data selection bias, in: *Proceedings of the Twenty-Ninth International Conference on Artificial Intelligence*, 2021, pp. 2177–2183.
- [4] Z. Shen, P. Cui, K. Kuang, B. Li, P. Chen, Causally regularized learning with agnostic data selection bias, in: *Proceedings of the 26th ACM international conference on Multimedia*, 2018, pp. 411–419.
- [5] F. Anowar, S. Sadaoui, B. Selim, Conceptual and empirical comparison of dimensionality reduction algorithms (pca, kpca, lda, mds, svd, lle, isomap, le, ica, t-sne), *Computer Science Review* 40 (2021).
- [6] N. Sharma, K. Saroha, Study of dimension reduction methodologies in data mining, in: *International Conference on Computing, Communication & Automation*, IEEE, 2015, pp. 133–137.
- [7] A.M. Martinez, A.C. Kak, Pca versus lda, *IEEE transactions on pattern analysis and machine intelligence* 23 (2) (2001) 228–233.
- [8] I. Jolliffe, Principal component analysis, *Encyclopedia of statistics in behavioral science* (2005).
- [9] H. Yu, J. Yang, A direct LDA algorithm for high-dimensional data—with application to face recognition, *Pattern recognition* 34 (10) (2001) 2067–2070.
- [10] S.T. Roweis, L.K. Saul, Nonlinear dimensionality reduction by locally linear embedding, *Science* 290 (5500) (2000) 2323–2326.
- [11] X. He, P. Niyogi, Locality preserving projections, *Advances in neural information processing systems* 16 (2003).
- [12] J.B. Tenenbaum, V.D. Silva, J.C. Langford, A global geometric framework for nonlinear dimensionality reduction, *Science* 290 (5500) (2000) 2319–2323.
- [13] M. Belkin, P. Niyogi, Laplacian eigenmaps and spectral techniques for embedding and clustering, *Advances in neural information processing systems* 14 (2001).
- [14] L. Cao, K.S. Chua, W. Chong, H. Lee, Q. Gu, A comparison of pca, kpca and ica for dimensionality reduction in support vector machine, *Neurocomputing* 55 (1–2) (2003) 321–336.
- [15] K.-R. Müller, S. Mika, K. Tsuda, K. Schölkopf, An introduction to kernel-based learning algorithms, in: *Handbook of neural network signal processing*, CRC Press, 2018, 1–4.
- [16] S. Yan, D. Xu, B. Zhang, H.-J. Zhang, Graph embedding: A general framework for dimensionality reduction, 2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05), Vol. 2, IEEE, 2005, pp. 830–837.
- [17] X. He, D. Cai, S. Yan, H.-J. Zhang, Neighborhood preserving embedding, *Tenth IEEE International Conference on Computer Vision (ICCV'05)*, Volume 1, Vol. 2, IEEE, 2005, pp. 1208–1213.
- [18] L. Qiao, S. Chen, X. Tan, Sparsity preserving projections with applications to face recognition, *Pattern Recognition* 43 (1) (2010) 331–341.
- [19] J. Yang, D. Zhang, J.-Y. Yang, B. Niu, Globally maximizing, locally minimizing: unsupervised discriminant projection with applications to face and palm biometrics, *IEEE transactions on pattern analysis and machine intelligence* 29 (4) (2007) 650–664.
- [20] S. Yan, D. Xu, B. Zhang, H.-J. Zhang, Q. Yang, S. Lin, Graph embedding and extensions: A general framework for dimensionality reduction, *IEEE transactions on pattern analysis and machine intelligence* 29 (1) (2006) 40–51.
- [21] P.R. Rosenbaum, D.B. Rubin, The central role of the propensity score in observational studies for causal effects, *Biometrika* 70 (1) (1983) 41–55.
- [22] E.A. Stuart, Matching methods for causal inference: A review and a look forward, *Statistical science: a review journal of the Institute of Mathematical Statistics* 25 (1) (2010) 1.
- [23] M. Lechner, Earnings and employment effects of continuous gff-the-job training in east germany after unification, *Journal of Business & Economic Statistics* 17 (1) (1999) 74–90.
- [24] M.J. Funk, D. Westreich, C. Wiesen, T. Stürmer, M.A. Brookhart, M. Davidian, Doubly robust estimation of causal effects, *American journal of epidemiology* 173 (7) (2011) 761–767.
- [25] J. Hainmueller, Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies, *Political analysis* 20 (1) (2012) 25–46.
- [26] S. Athey, G.W. Imbens, S. Wager, Approximate residual balancing: debiased inference of average treatment effects in high dimensions, *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 80 (4) (2018) 597–623.
- [27] K. Kuang, P. Cui, B. Li, M. Jiang, S. Yang, Estimating treatment effect in the wild via differentiated confounder balancing, in: *Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining*, 2017, pp. 265–274.
- [28] D. Cai, X. He, J. Han, Spectral regression: A unified approach for sparse subspace learning, in: *Seventh IEEE international conference on data mining (ICDM 2007)*, IEEE, 2007, pp. 73–82.
- [29] X. He, D. Cai, S. Yan, H.-J. Zhang, Neighborhood preserving embedding, *Tenth IEEE International Conference on Computer Vision (ICCV'05)*, Volume 1, Vol. 2, IEEE, 2005, pp. 1208–1213.
- [30] D. Cai, X. He, J. Han, Spectral regression for efficient regularized subspace learning, in: *2007 IEEE 11th international conference on computer vision*, IEEE, 2007, pp. 1–8.
- [31] K. Kuang, P. Cui, S. Athey, R. Xiong, B. Li, Stable prediction across unknown environments, in: *proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining*, 2018, pp. 1617–1626.
- [32] M. Long, J. Wang, G. Ding, J. Sun, P. S. Yu, Transfer joint matching for unsupervised domain adaptation, in: *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2014, pp. 1410–1417.

- [33] B. Gong, Y. Shi, F. Sha, K. Grauman, Geodesic flow kernel for unsupervised domain adaptation, in: 2012 IEEE conference on computer vision and pattern recognition, 2012, pp. 2066–2073.
- [34] H. Venkateswara, J. Eusebio, S. Chakraborty, S. Panchanathan, Deep hashing network for unsupervised domain adaptation, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2017, pp. 5018–5027.
- [35] H. Bay, T. Tuytelaars, L.V. Gool, Surf: Speeded up robust features, in: European conference on computer vision, Springer, 2006, pp. 404–417.
- [36] L. Van der Maaten, A new benchmark dataset for handwritten character recognition, Tilburg University (2009) 2–5.
- [37] M. Balasubramanian, E.L. Schwartz, The isomap algorithm and topological stability, *Science* 295 (5552) (2002), 7–7.
- [38] J. Wang, F. Xie, F. Nie, X. Li, Unsupervised adaptive embedding for dimensionality reduction, *IEEE Transactions on Neural Networks and Learning Systems* 33 (11) (2021) 6844–6855.



Xiuqi Huang received the B.S. degree in Statistics from Huazhong University of Science and Technology, Wuhan, China. He is currently pursuing a master's degree in System Science from the National University of Defense Technology, Changsha, China. His research interests include machine learning and causal inference.



Haotian Ni is a Ph.D. candidate at the National University of Defense Technology, Changsha, China. He received the B.S. degree from Central South University, Changsha, China, in 2017 and the M.S. degree from Academy of Mathematics and Systems Science Chinese Academy of Sciences, Beijing, China in 2020. His research interests include causal inference, machine learning, and data mining.



Tingjin Luo received his Ph.D degrees from the National University of Defense Technology. He is currently an associate Professor with the College of Science of the same university. He was a visiting Ph.D student with the University of Michigan, Ann Arbor, USA, from 2015 to 2017. He has authored more than 40 papers in journals and conferences, such as the IEEE TKDE, IEEE TCYB, ACM KDD and ICME etc. He has been a Program Committee member of several conferences including IJCAI, AAAI and ICPR etc. His research interests include machine learning, optimization, data mining and computer vision.



Hong Tao received the Ph.D. degree from the National University of Defense Technology, Changsha, China, in 2019. She is currently an Associate Professor with the College of Science of the same university. Her research interests include machine learning, system science, and data mining.



Chenping Hou received the B.S. and Ph.D. degrees in applied mathematics from the National University of Defense Technology, Changsha, China, in 2004 and 2009, respectively. He is currently a full Professor with the College of Science of the same university. He has authored more than 100 papers in journals and conferences, such as the IEEE TPAMI, IEEE TKDE, IEEE TIP, IJCAI, AAAI, etc. His current research interests include pattern recognition, machine learning, data mining, and computer vision.